

EMILIA Protocol — the accountability layer for the agent economy

Portable, offline-verifiable proof that a named human authorized each irreversible action an AI agent takes.

For fifty years, security answered "*who is allowed in?*" The dominant users of software are now AI agents that move money, change records, and delete data on the fly. The unanswered question is **authorization at the moment of action: who approved *that exact action*, before it executed?** "A human was in the loop" is, today, an unfalsifiable claim — and regulators (EU AI Act Art. 14) now require oversight no one can verify after the fact.

WHAT EP IS

Before a high-risk action runs, a **named human** approves the exact action on their own device (WebAuthn / Face ID). EP emits a signed **authorization receipt** anyone can verify **offline, no account, without trusting our servers** — and it stays valid even if EP disappears. *Decision logs are testimony; EP produces receipts.*

WHY IT'S REAL, AND CHECKABLE

- **3 languages / 8 suites:** JS, Python, Go verifiers agree on 8 adversarial cross-language conformance suites — the IETF bar for a standard.
- **Independently verified (2026):** an outside implementer ran the crash-test + conformance on their own machine and confirmed it — the first external reliance event.
- **Formal + standardized:** machine-checked TLA+ & Alloy in CI; authorization-receipt and quorum Internet-Drafts posted at the IETF.

THE CATEGORY JUST GOT FUNDED — WE'RE THE LAYER UNDERNEATH

2026 confirmed agent-action security is a category: **Arcade \$60M** (SYN Ventures, Morgan Stanley), **Oasis \$120M** (Craft, Sequoia, Accel), **GitGuardian \$50M** (Insight), Astrix acquired by Cisco. Identity/NHI proves *which machine* acts; action-auth proves *which tools* it may call. EP proves the piece none of them do — **which named human authorized this exact action**, verifiable offline. Complement, not competitor: the accountability artifact regulators demand and the funded stack doesn't emit.

WEDGE: PAYMENT-DESTINATION FRAUD

Vendor bank-detail changes & beneficiary updates are where valid access turns into real loss (BEC). EP requires a named human to approve the exact change before money moves, then exports an evidence packet an auditor or insurer can verify. Same receipt covers clinical double-checks & defense human-control.

MODEL: OPEN CORE

Format & verifier are free (ubiquity); the **operated trust root** — directory, transparency log, revocation, evidence pipelines — is the revenue layer an org can't casually self-host.

HOW IT SHIPS

Apache-2.0 + a one-file **zero-dependency drop-in** + npm; kits for MCP, Vercel Eve, and agent runtimes; IETF drafts that compose with **SCITT (RFC 9943)**. Emitted by every platform, owned by none.

THE RAISE

\$2–3M seed: team (GTM lead, AI-protocol & network engineers, a standards & policy lead — adoption, lobbying, testing), adoption & standards, and the managed control plane to first regulated-deployer revenue.

WHY INEVITABLE

Public-key crypto existed two decades before the browser padlock made its *absence* unacceptable. EP becomes obvious the same way — an insurer's coverage condition, an auditor's framework, or a government AI-auditor registry making the **absence of a receipt** unacceptable. Precedent → recommendation → requirement.

Open standard (Apache-2.0) · IETF Internet-Drafts (composes with SCITT / RFC 9943) · 3 verifiers / 8 conformance suites · RR-1 conformance · zero-dependency drop-in · TLA+ & Alloy in CI · EU AI Act / DoD crosswalk · try it offline: `npx @emilia-protocol/crash-test`

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